Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

SDC HACKATHON 2026

Phase 1: Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

1. Team & Project Overview

Team Name	K&Z
Selected Domain	Domain 1
Selected SDG Track (1–17)	SDG 4, 8, 9
Project / Solution Name	K&Z
Name (Leader)	Chong Wen Kai
Email & Phone (Leader)	chongwenkai0@gmail.com (0123909932)
University Affiliation(s)	Xiamen University Malaysia
Number of Team Members	2

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

AI has been rapidly transforming the workforce, changing the skills employers need and types of jobs available. In Malaysia, this creates challenges for working adults facing potential job displacement, fresh graduates entering a competitive job market, career switchers exploring new job opportunities and employers seeking to prepare their workforce for technological change.

However, many individuals lack of clear guidance on which skills to develop, which careers offer promising future opportunities and how to transition into emerging roles. Existing career guidance often focuses on current qualification and job requirements, while overlooking evolving AI trends, personalized reskilling pathways and long term relevance of different skills. Consequently, individuals risk investing time, and money in skills that may become less relevant while employers face difficulties identifying and addressing workforce skill gaps.

These challenges also have wider economic and social implications. Skills mismatches can contribute to unemployment, unequal access to career opportunities, and reduced productivity. Meanwhile, the potential of future careers to support sustainable development is often overlooked, including contributions to quality education, decent work, industrial innovation, and climate action.

To address these problems, K&Z proposes an AI-powered career intelligence platform that connects personal skills and interests with future career opportunities. By combining AI-trend signals, career matching, skills-gap analysis, personalised reskilling recommendations, and Sustainable Development Goal (SDG) mapping. We also aims to help individuals make informed, future-ready career decision while supporting a more inclusive, productive and sustainable workforce.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

K&Z proposes an AI-powered career intelligence and psychological career counselling platform designed to help individuals navigate an evolving job market with greater confidence and clarity. Unlike conventional career platforms that focus primarily on qualifications and job matching, K&Z integrates AI-driven career recommendations with psychological career counselling to consider users' interests, strengths, confidence, motivations, and career concerns.

The application combines resume analysis, RIASEC-based career assessment and labour market intelligence to identify suitable future career opportunities. By leveraging generative AI APIs can interpret AI and job-market trends, explain emerging skills and provide personalised career insights. Its skills-gap analysis identifies areas for improvement, while its reskilling module recommends structured learning pathways to help users prepare for future roles. We also added psychological career counselling complements these features by supporting career self-awareness, decision-making confidence and reflection on personal values and career uncertainty, ensuring that recommendation consider both employability and individual well-being.

In addition, we incorporates SDG mapping to demonstrate how recommended career and skills can contribute to goals such as quality education, decent work, industry innovation and climate action.

For development and deployment, K&Z intends to leverage knowledge gained form SDC phase 1 workshops on Runpod and Replit. Replit will support application development and deployment, while Runpod will provide a platform for hosting and running computationally demanding AI models. The application also uses FastAPI, python-based machine learning models and structured datasets covering occupations, skills, and Malaysia labour-market information.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Component	Technology	Purpose
Frontend	HTML, CSS , Javascript	Build user interface for career assessments, career matching, SDG impact and reskilling.
Backend	Python, FastAPI	Handle application routes, user requests, AI processing and communication between modules
Generative AI API	Deepseek API	Generate personalized career insights, explain skills gaps and support psychological career reflection
Machine learning	Distilbert	Analyse resume and support career classification and matching
Data source	DOSM	Provide Malaysia labour-market and wage data for career and salary insights
Model hosting	Runpod	Host and run computationally demanding AI models
Application deployment	Replit	Support development, testing and deployment web application

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	According to the Department of Statistics Malaysia (DOSM), Malaysia recorded a 32.2% skill-related underemployment rate among employed graduates in 2024, affecting approximately 1.60 million graduates. This indicates a continuing challenge in aligning graduates' qualifications and skills with the requirements of their employment. This national statistic provides the problem baseline for K&Z. It is not a direct measurement of the platform's impact and the prototype is not expected to change the national rate within the hackathon period.
Core Measurable KPI *The specific unit you will track*	Percentage of test users who successfully complete the career-readiness journey, including receiving a career recommendation, identifying at least one relevant skills gap, and obtaining an actionable reskilling pathway.

Metric Category	Your Response
e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.	The team will measure task completion using a controlled prototype evaluation with synthetic career profiles and volunteer testers. The evaluation will also assess whether users understand the recommendations and their supporting evidence.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	By the final prototype presentation, K&Z aims to achieve at least 80% successful completion of career-readiness journey among a minimum of 10 test profiles and participants. Our team will record the completion rates, recommendation relevance, and user feedback to evaluate the prototype's feasibility and usefulness. Results will be reported as prototype evaluation outcomes rather than evidence of a reduction in Malaysia's national underemployment rate.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ ✓ Yes ☐ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

Chong Wen Kai (resume analyses distilbert)

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☐ ✓ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Department of statistics Malaysia (DOSM)	**dataset**	**verify the applicable dataset terms of use**
O*NET database (RIASEC)	**dataset**	**O*NET Data license**

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☐✓ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[]

[23/10/2026]